



Mandarin Phonetics Representations Learning via BiLSTM

Ilya Grigorev

il.grigorev@innopolis.university
Innopolis University

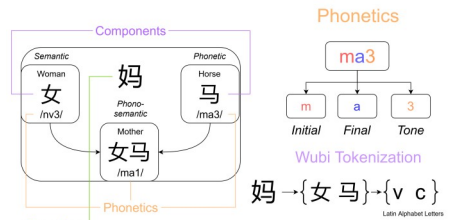


PAPER & CODE

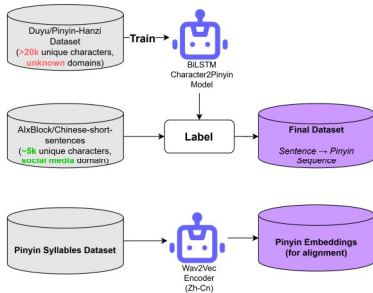
Motivation

Many representation-learning tasks aim for embeddings whose geometry reflects similarity along a target attribute — phonetic, semantic, or visual — not just accuracy on a discriminative loss. Yet standard supervised training (cross-entropy over discrete labels) puts no direct pressure on the hidden space: a model can reach near-perfect label accuracy while embedding near-identical inputs far apart. We study this in Mandarin phonetics, where every character carries a well-defined pronunciation and a rich network of homophones that stresses any representation claiming to encode sound. On a BiLSTM encoder-decoder we test three explicit strategies for shaping the hidden geometry: a shared sub-character input basis (Wubi radical decomposition), a continuous acoustic target space (wav2vec2 pinyin embeddings) the hidden states are aligned to, and dimensionality reduction to expose latent structure. To measure each, we design an evaluation framework that decouples the visual, phonetic, acoustic, and semantic contributions to hidden-space similarity, using a decoupling regression and decoder perturbation probes.

Experimental Setting



Dataset Collection



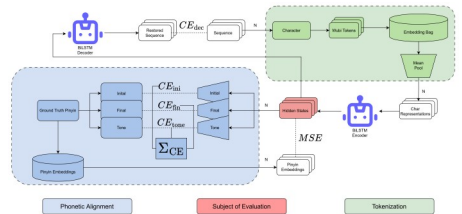
Loss Functions

$$\mathcal{L}_{\text{pinyin}} = \text{CE}_{\text{dec}} + \lambda_{\text{enc}} \Sigma_{\text{CE}}$$

$$\mathcal{L}_{\text{aligned-pinyin}} = \text{CE}_{\text{dec}} + \lambda_{\text{enc}} (\Sigma_{\text{CE}} + \lambda_{\text{MSE}} \text{MSE})$$

$$\mathcal{L}_{\text{aligned}} = \text{CE}_{\text{dec}} + \lambda_{\text{enc}} \text{MSE}$$

Training Pipeline



Final Effect Estimation

Pinyin Encoder (No alignment) — $R^2 = 0.268$

$$\sigma(\mathbf{h}_i, \mathbf{h}_j) = 0.61\sigma(\mathbf{s}_i, \mathbf{s}_j) + 0.11\sigma(\mathbf{p}_i, \mathbf{p}_j) + 0.06\sigma(\mathbf{w}_i, \mathbf{w}_j) + 0.06\sigma(\mathbf{p}_i, \mathbf{p}_j) + 0.03\sigma(\mathbf{w}_i, \mathbf{w}_j) + 0.63f(\mathbf{w}_i, \mathbf{w}_j) + 0.001\sigma(\mathbf{v}_i, \mathbf{v}_j) - 0.01$$

Aligned Pinyin Encoder (with alignment) — $R^2 = 0.611$

$$\sigma(\mathbf{h}_i, \mathbf{h}_j) = 0.16\sigma(\mathbf{s}_i, \mathbf{s}_j) + 0.15\sigma(\mathbf{p}_i, \mathbf{p}_j) + 0.12\sigma(\mathbf{w}_i, \mathbf{w}_j) + 0.10\sigma(\mathbf{p}_i, \mathbf{p}_j) + 0.07\sigma(\mathbf{w}_i, \mathbf{w}_j) + 0.61f(\mathbf{w}_i, \mathbf{w}_j) + 0.004\sigma(\mathbf{v}_i, \mathbf{v}_j) + 0.04$$

Aligned Encoder (alignment only) — $R^2 = 0.805$

$$\sigma(\mathbf{h}_i, \mathbf{h}_j) = 0.26\sigma(\mathbf{s}_i, \mathbf{s}_j) - 0.06\sigma(\mathbf{p}_i, \mathbf{p}_j) + 0.002\sigma(\mathbf{w}_i, \mathbf{w}_j) - 0.03\sigma(\mathbf{p}_i, \mathbf{p}_j) - 0.03\sigma(\mathbf{w}_i, \mathbf{w}_j) + 0.004f(\mathbf{w}_i, \mathbf{w}_j) + 0.003\sigma(\mathbf{v}_i, \mathbf{v}_j) + 0.18$$

$\sigma(\cdot, \cdot)$ — cosine similarity

$\mathbb{I}_{\text{wubi}}(\cdot, \cdot)$ — pinyins share a trait

$J(\cdot, \cdot)$ — Jaccard Similarity

$\mathbf{h}$ — hidden state vector

$\mathbf{w}$ — wubi tokens set

$\mathbf{p}$ — pinyin decomposition

$\mathbf{v}$ — semantic embedding (BERT)

$\mathbf{a}$ — acoustic vector (pinyin embedding)

Conclusions

Phonetic alignment produces sufficient effect towards expected contextual embeddings for characters

Dimensionality reduction resulted in sufficient improvements in recovering the similarity sources

Wubi tokenization acts as a grouping basis for characters